

# Software Testing Document SweetLedger



Version 1.5 (Updated as of 12/07/25)

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# **1. Introduction**

This document defines the test cases for verifying the functional and non-functional requirements of the SweetLedger Accounting & Financial Management System. Each requirement is mapped to a test case for full traceability.

## **2. Test Objectives**

Through these tests, the SweetLedger team looks to validate all system functionalities, ensure role-based access control, confirm authentication, login, and password rules, and ensure UI consistency and ease of use.

### **2.1 Test Selection**

Unit tests and end-to-end tests which work at a lower level to validate operation of more granular pieces of code. System integration and end-to-end will be finalized as the project comes to its end. Each test will run across the pages and the functions required for operation of each page separately to be confirmed. Playwright tests are used to simulate user experience, whereas vitest is being used to test the functions and the range across their use.

### **2.2 Adequacy Criterion**

Most of the current testing relies on expected outcomes to the functions and defining where failure will occur as well as how the application will handle invalid inputs. This

helps to ensure that failures taken in will be handled in an appropriate manner which will never break the application nor halt runtimes.

### 2.3 Bug Tracking

The testing suite proactively detects updates in the code and because of the structure of the tests, the bugs can be tracked directly to the component as well as the function which failed the test in case of any change within the codebase. Any significant bugs are tracked through the trello and a subsequent test is created to ensure that the bug case can never occur.

### 2.4 Technology and Tools

Vitest and Playwright are the automated testing suites being used for testing the application because it integrates directly with the project, as it is built in React using Vite as a build tool. JSDom is utilized to ensure that browser operations are consistent like deployed builds. It launches a browser instance and controls the DOM with operations which mimic a user.

## 3. Test Scope

- **Functional Testing:** Login, Event Log, User Management, Chart of Accounts, Journal, Ledger, Reports.
- **Security Testing:** Authentication, password requirements, role-based access control.
- **UI Testing:** Layout consistency, error messages, dashboards.

- **Database Connection Testing:** Posting new data, updating new data, and event logging

## 4. Test Case Table

### 4.1 Account & User Management (ACM, ADM)

| Test Case ID | Req ID | Description                                              | Preconditions       | Test Steps                                                                               | Expected Result                                                | Actual Result                                                  | P/F |
|--------------|--------|----------------------------------------------------------|---------------------|------------------------------------------------------------------------------------------|----------------------------------------------------------------|----------------------------------------------------------------|-----|
| TC-ACM-01    | ACM-01 | Verify that Admins, Accountants, and Managers can log in | User accounts exist | 1. Navigate to login page<br>2. Enter valid credentials for each role<br>3. Click submit | User is successfully logged in and redirected to the dashboard | User is successfully logged in and redirected to the dashboard | P   |
| TC-ACM-02    | ACM-02 | Verify role-based dashboard redirection                  | User accounts exist | 1. Login as Admin/Manager/Accountant<br>2. Verify dashboard view                         | Dashboard corresponds to user role                             | Dashboard corresponds to user role                             | P   |
| TC-ACM-03    | ACM-03 | Verify new user registration request                     | None                | 1. Navigate to Create New User                                                           | Request submitted,                                             | Request submitted,                                             | F   |

|           |        |                                                               |                             |                                                    |                                      |                                  |   |
|-----------|--------|---------------------------------------------------------------|-----------------------------|----------------------------------------------------|--------------------------------------|----------------------------------|---|
|           |        |                                                               |                             | 2. Enter valid details<br>3. Submit request        | visible to Admin                     | not visible to admin             |   |
| TC-ACM-04 | ACM-04 | Verify Admin receives email to approve new user               | Registration request exists | 1. Submit new user request<br>2. Check Admin email | Admin receives approval notification | Approval notification not sent   | F |
| TC-ACM-05 | ACM-05 | Verify username pattern<br><FirstInitial><FullLastName><MMYY> | None                        | 1. Attempt to create username not matching pattern | Auto Generate Read only username     | Auto Generate Read only username | P |
| TC-ADM-01 | ADM-01 | Verify Admin can access all system user reports               | Admin logged in             | 1. Navigate to Reports page<br>2. View all users   | All users displayed accurately       | All users displayed accurately   | P |
| TC-ADM-02 | ADM-02 | Verify Admin can suspend a user                               | Admin logged in             | 1. Select user                                     | User suspended for date range        | User suspended                   | P |

|           |        |                                      |                 |                                 |                 |                 |   |
|-----------|--------|--------------------------------------|-----------------|---------------------------------|-----------------|-----------------|---|
|           |        |                                      |                 | 2. Set suspension dates         |                 | for date range  |   |
| TC-ADM-03 | ADM-03 | Verify Admin can send email to users | Admin logged in | 1. Select user<br>2. Send email | Email delivered | Email delivered | P |

#### 4.2 Chart of Accounts (COA)

| Test Case ID | ReqID  | Description                                           | Preconditions   | Test Steps                         | Expected Result               | Actual Result                 | P/F |
|--------------|--------|-------------------------------------------------------|-----------------|------------------------------------|-------------------------------|-------------------------------|-----|
| TC-COA-01    | COA-01 | Verify Admin can add, edit, view, deactivate accounts | Admin logged in | 1.Add/Edit/View/Deactivate account | Account changes successful    | Account changes successful    | P   |
| TC-COA-02    | COA-02 | Verify all required fields are included in account    | Admin logged in | 1. Add account with all fields     | Account saved with all fields | Account saved with all fields | P   |

|           |        |                                                         |                 |                                               |                                      |                                      |   |
|-----------|--------|---------------------------------------------------------|-----------------|-----------------------------------------------|--------------------------------------|--------------------------------------|---|
| TC-COA-03 | COA-03 | Verify no duplicate account names or numbers            | Admin logged in | 1. Add duplicate account                      | System prevents duplicates           | System prevents duplicates           | P |
| TC-COA-04 | COA-04 | Verify monetary values formatting                       | Admin logged in | 1. Add account with value 20                  | Displayed as 20.00                   | Displayed as 20.00                   | P |
| TC-COA-05 | COA-05 | Verify accounts with balance > 0 cannot be deactivated  | Admin logged in | 1. Attempt to deactivate account with balance | Deactivation prevented               | Deactivation not enabled             | F |
| TC-COA-06 | COA-06 | Verify Admin can view account details & generate report | Admin logged in | 1. View account details<br>2. Generate report | Details displayed & report generated | Details displayed & report generated | P |
| TC-COA-07 | COA-07 | Verify search & filter accounts                         | Admin logged in | 1. Search/filter accounts                     | Correct results displayed            | Correct results displayed            | P |



|           |        |                                                      |                              |                                 |                         |                         |   |
|-----------|--------|------------------------------------------------------|------------------------------|---------------------------------|-------------------------|-------------------------|---|
| TC-COA-08 | COA-08 | Verify account click navigates to ledger             | Manager/Accountant logged in | 1. Click account name/number    | Navigate to ledger page | Navigate to ledger page | P |
| TC-COA-09 | COA-09 | Verify Managers & Accountants cannot modify accounts | Manager/Accountant logged in | 1. Attempt add/edit/deactivate  | Action not provided     | Action not provided     | P |
| TC-COA-10 | COA-10 | Verify users can send emails from COA page           | User logged in               | 1. Select user<br>2. Send email | Email delivered         | Email delivered         | P |

#### 4.3 Database & Event Logs (DBM, EVL, ERR)

| Test Case ID | Req ID | Description                                                           | Preconditions | Test Steps                 | Expected Result           | Actual Result              | P/F       |
|--------------|--------|-----------------------------------------------------------------------|---------------|----------------------------|---------------------------|----------------------------|-----------|
| TC-DBM-01    | DBM-01 | Verify all login attempts, password histories, user activities stored | Users exist   | 1. Perform login & actions | All events recorded in DB | Some events recorded in DB | Partial P |

|           |        |                                                          |                         |                                            |                         |                         |   |
|-----------|--------|----------------------------------------------------------|-------------------------|--------------------------------------------|-------------------------|-------------------------|---|
| TC-DBM-02 | DBM-02 | Verify event logs for each record stored                 | Account exists          | 1. Add/Edit/Deactivate account             | Event log created in DB | Event log created in DB | P |
| TC-EVL-01 | EVL-01 | Verify event log created for account actions             | Account exists          | 1. Add/Edit/Deactivate account             | Event logged            | Event logged            | P |
| TC-EVL-02 | EVL-02 | Verify logs show before/after images, user ID, timestamp | Account exists          | 1. Make changes to account<br>2. View logs | Logs display all info   | Logs display all info   | P |
| TC-ERR-01 | ERR-01 | Verify error messages displayed in red                   | System generates errors | 1. Trigger error                           | Message shown in red    | Message shown in red    | P |
| TC-ERR-02 | ERR-02 | Verify all errors logged in DB                           | System generates errors | 1. Trigger error                           | Error recorded in DB    | Error recorded in DB    | P |

#### 4.4 Dashboard & Financials (DSH, FIN)

| Test Case ID | Req ID | Description                                | Preconditions  | Test Steps                      | Expected Result                | Actual Result                  | P/F |
|--------------|--------|--------------------------------------------|----------------|---------------------------------|--------------------------------|--------------------------------|-----|
| TC-DSH-01    | DSH-01 | Verify landing page as dashboard           | User logged in | 1. Login<br>2. Verify dashboard | Dashboard displayed            | Dashboard displayed            | P   |
| TC-DSH-02    | DSH-02 | Verify menu buttons correspond to role     | User logged in | 1. View dashboard buttons       | Buttons match role permissions | Buttons match role permissions | P   |
| TC-DSH-03    | DSH-03 | Verify important system messages displayed | User logged in | 1. Check message section        | Messages visible               | Messages visible               | P   |
| TC-DSH-04    | DSH-04 | Verify color scheme consistent             | User logged in | 1. View dashboard               | Colors match application       | Colors match application       | P   |
| TC-FIN-01    | FIN-01 | Verify financial ratios visible            | User logged in | 1. Check dashboard ratios       | Ratios displayed               | Ratios displayed               | P   |

|           |        |                                      |                |                 |                                |                                |   |
|-----------|--------|--------------------------------------|----------------|-----------------|--------------------------------|--------------------------------|---|
| TC-FIN-02 | FIN-02 | Verify three-color scheme for ratios | User logged in | 1. Check ratios | Ratios show green, yellow, red | Ratios show green, yellow, red | P |
|-----------|--------|--------------------------------------|----------------|-----------------|--------------------------------|--------------------------------|---|

#### 4.5 Journal (JOR)

| Test Case ID | Req ID | Description                                           | Preconditions      | Test Steps                                        | Expected Result | Actual Result  | P/F |
|--------------|--------|-------------------------------------------------------|--------------------|---------------------------------------------------|-----------------|----------------|-----|
| TC-JOR-01    | JOR-01 | Verify creation of journal entries using COA          | COA accounts exist | 1. Create new journal entry<br>2. Select accounts | Entry accepted  | Entry accepted | P   |
| TC-JOR-02    | JOR-02 | Verify debit before credit & multiple entries allowed | Journal exists     | 1. Create entry with multiple debits & credits    | Entry accepted  | Entry accepted | P   |
| TC-JOR-03    | JOR-03 | Verify reset/cancel                                   | Entry created      | 1. Start entry<br>2. Reset/cancel                 | Entry cleared   | Entry cleared  | P   |

|           |        |                                           |                 |                                               |                                |                                |   |
|-----------|--------|-------------------------------------------|-----------------|-----------------------------------------------|--------------------------------|--------------------------------|---|
|           |        | allowed before submission                 |                 |                                               |                                |                                |   |
| TC-JOR-04 | JOR-04 | Verify at least one debit & credit        | Entry created   | 1. Submit entry without debit or credit       | Submission rejected            | Submission rejected            | P |
| TC-JOR-05 | JOR-05 | Verify debits equal credits               | Entry created   | 1. Submit entry                               | System validates equality      | System validates equality      | P |
| TC-JOR-06 | JOR-06 | Verify source document attachment         | Entry created   | 1. Attach PDF/Word/Excel/CSV/JPG/PNG          | Attachment successful          | Attachment successful          | P |
| TC-JOR-07 | JOR-07 | Verify manager approval/rejection         | Entry submitted | 1. Approve/Reject entry<br>2. Provide comment | Status updated & comment saved | Status updated & comment saved | P |
| TC-JOR-08 | JOR-08 | Verify approved entry reflected in ledger | Entry approved  | 1. Check ledger                               | Ledger updated                 | Ledger updated                 | P |

|           |        |                                                         |                 |                         |                               |                               |   |
|-----------|--------|---------------------------------------------------------|-----------------|-------------------------|-------------------------------|-------------------------------|---|
| TC-JOR-09 | JOR-09 | Verify filter entries by date/status                    | Entries exist   | 1. Filter entries       | Correct entries displayed     | Correct entries displayed     | P |
| TC-JOR-10 | JOR-10 | Verify viewing entries by other managers/accountholders | Entries exist   | 1. View entries         | Entries visible               | Entries visible               | P |
| TC-JOR-11 | JOR-11 | Verify invalid entries cannot be submitted              | Entry created   | 1. Submit invalid entry | Submission blocked            | Submission blocked            | P |
| TC-JOR-12 | JOR-12 | Verify system notification on submission                | Entry submitted | 1. Submit entry         | Managers receive notification | Managers receive notification | P |

#### 4.6 Ledger (LED)

| Test Case ID | Req ID | Description | Preconditions | Test Steps | Expected Result | Actual Result | P/F |
|--------------|--------|-------------|---------------|------------|-----------------|---------------|-----|
|--------------|--------|-------------|---------------|------------|-----------------|---------------|-----|

|           |        |                                                    |               |                               |                                            |                                            |   |
|-----------|--------|----------------------------------------------------|---------------|-------------------------------|--------------------------------------------|--------------------------------------------|---|
| TC-LED-01 | LED-01 | Verify clicking account number navigates to ledger | Entry exists  | 1. Click account number       | Navigate to ledger page                    | Navigate to ledger page                    | P |
| TC-LED-02 | LED-02 | Verify ledger displays all transaction info        | Ledger exists | 1. View ledger                | Date, desc, debit, credit, balance visible | Date, desc, debit, credit, balance visible | P |
| TC-LED-03 | LED-03 | Verify PR clickable to journal entry               | Ledger exists | 1. Click PR                   | Navigate to journal entry                  | Navigate to journal entry                  | P |
| TC-LED-04 | LED-04 | Verify filtering ledger entries                    | Ledger exists | 1. Filter by date/name/amount | Filtered results shown                     | Filtered results shown                     | P |
| TC-LED-05 | LED-05 | Verify ledger balances accurate                    | Ledger exists | 1. Post entries               | Balances correct                           | Balances correct                           | P |

|  |  |  |  |                       |  |  |  |
|--|--|--|--|-----------------------|--|--|--|
|  |  |  |  | 2. Verify<br>balances |  |  |  |
|--|--|--|--|-----------------------|--|--|--|

#### 4.7 Reports (REP)

| Test<br>Case ID | Req ID | Description                                                                                              | Preconditions  | Test Steps                                                        | Expected<br>Result                    | Actual<br>Result                      | P/F |
|-----------------|--------|----------------------------------------------------------------------------------------------------------|----------------|-------------------------------------------------------------------|---------------------------------------|---------------------------------------|-----|
| TC-REP<br>-01   | REP-01 | Verify Trial<br>Balance,<br>Income<br>Statement,<br>Balance<br>Sheet,<br>Retained<br>Earnings<br>reports | Accounts exist | 1. Generate<br>reports<br><br>2.<br><br>View/Save/E<br>mail/Print | Reports<br>generated<br><br>correctly | Reports<br>generated<br><br>correctly | P   |

#### 4.8 Security & Login (SEC, UI)

| Test<br>Case ID | Req ID | Description | Preconditions | Test Steps | Expected Result | Actual<br>Result | P/F |
|-----------------|--------|-------------|---------------|------------|-----------------|------------------|-----|
|-----------------|--------|-------------|---------------|------------|-----------------|------------------|-----|



|           |        |                                                        |             |                                   |                                   |                               |   |
|-----------|--------|--------------------------------------------------------|-------------|-----------------------------------|-----------------------------------|-------------------------------|---|
| TC-SEC-01 | SEC-01 | Verify login using DB credentials                      | User exists | 1. Login with valid credentials   | Login successful                  | Login successful              | P |
| TC-SEC-02 | SEC-02 | Verify password complexity                             | User exists | 1. Set password not meeting rules | System rejects password           | System rejects password       | P |
| TC-SEC-03 | SEC-03 | Verify role-based page access                          | User exists | 1. Access restricted page         | Access denied if unauthorized     | Access denied if unauthorized | P |
| TC-SEC-04 | SEC-04 | Verify auto-suspension after 3 failed logins           | User exists | 1. Attempt 3 failed logins        | Account suspended                 | Account suspended             | P |
| TC-SEC-05 | SEC-05 | Verify password expiration notifications               | User exists | 1. Login near password expiration | Notification sent & reuse blocked | Not implemented               | F |
| TC-SEC-06 | SEC-06 | Verify reset password with email/ID/security questions | User exists | 1. Reset password                 | Reset successful                  | Reset successful              | P |

|          |       |                                                       |                |                                |                                         |                                         |   |
|----------|-------|-------------------------------------------------------|----------------|--------------------------------|-----------------------------------------|-----------------------------------------|---|
| TC-UI-01 | UI-01 | Verify login page fields & logo                       | None           | 1. Navigate to login page      | All fields & logo visible               | All fields & logo visible               | P |
| TC-UI-02 | UI-02 | Verify username/profile appears top-right after login | User logged in | 1. Login<br>2. Check top-right | Username/profile visible                | Username/profile visible                | P |
| TC-UI-03 | UI-03 | Verify UI consistency on all pages                    | User logged in | 1. Navigate pages              | Layout, logo, tooltips consistent       | Layout, logo, tooltips consistent       | P |
| TC-UI-04 | UI-04 | Verify calendar popup & navigation buttons            | User logged in | 1. Navigate pages              | Calendar & navigation buttons displayed | Calendar & navigation buttons displayed | P |

## 4.9 Navigation & Rendering

| Test Case | Purpose                    | Steps                                                                                       | Expected Result                                                                        | Actual Result                                                                      | Test Result |
|-----------|----------------------------|---------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------|------------------------------------------------------------------------------------|-------------|
| Login-001 | Renders default login page | Render <App/> then expect the frontpage text “SweetLedger” and sign in button to be visible | Initialized page, then successfully finding the login page text and the sign in button | The text element and the sign in button element are found and visible upon render. | Test Passes |

|                    |                                                |                                                                                                               |                                                                                                                                                      |                                                                            |             |
|--------------------|------------------------------------------------|---------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------|-------------|
| Login-002          | Invalid Login test for both empty fields       | The sign in button is clicked with no values in the username and password boxes                               | There must be text at the bottom of the page saying "Please enter both fields."                                                                      | The page returns red text saying 'Please enter both fields.'               | Test Passes |
| Login-003          | Invalid Login test with a short username       | The username and password boxes are given the value 'bad', then the sign in button is clicked                 | Text at the bottom of the page returned 'Username must be at least 8 characters.'                                                                    | The page returns red text saying 'Username must be at least 8 characters.' | Test Passes |
| Login-004          | Invalid login no credentials                   | The login function being tested when the input has no profile match                                           | Input 'badusername' to username and 'bad' to the password and checking for response                                                                  | The page returns red text saying 'Failed to load users.'                   | Test Passes |
| Login-005          | Identify Key Text elements on the login screen | Test checks that all text elements are present and named for the login page                                   | Check for Heading of name 'SweetLedger' and paragraph as subtitle 'Account Management System'. Username and Password Labels are checked for presence | All 4 text elements are found and rendered on the page as appropriate      | Test Passes |
| Login-006          | Expects the button elements on login page      | All 3 button elements, 'Sign In', 'Create New Account', and 'Clear' are checked by role and name for presence | All Buttons are returned, matching role and name                                                                                                     | All buttons return as expected by role and name                            | Test Passes |
| Login-007          | Expects the Link element in the login page     | Simple test for link role with the name 'Forgot Password?' expected as visible on screen                      | The Link is matched to the name and returned                                                                                                         | The link is present and has the name 'Forgot Password?'                    | Test Passes |
| Login-008          | Expect Image element                           | Expects the presence of an image element as well as an element with matching altText 'SweetLedger Logo'       | The screen finds the image element with the matching Alttext                                                                                         | The Image element returns with altText 'SweetLedger Logo'                  | Test Passes |
| ForgotPas<br>s-001 | Renders forgot                                 | The page is rendered, and the test expects an element unique to                                               | The page is confirmed to have                                                                                                                        | Render returned and element with 'Reset Your                               | Test Passes |

|                    |                                        |                                                                                                                                                                                                 |                                                                                                                                                |                                                                       |             |
|--------------------|----------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------|-------------|
|                    | password page                          | the page, text with 'Reset Your Password - Step 1 of 3'                                                                                                                                         | rendered with relevant elements                                                                                                                | Password - Step 1 of 3' found on the screen                           |             |
| ForgotPas<br>s-002 | Render initial text elements           | Confirms heading, username, and email text elements on initial render of the page.                                                                                                              | Screen checks for the heading, username label, and email label on screen and confirms their render in the page                                 | The test confirms the presence of all text elements                   | Test Passes |
| ForgotPas<br>s-003 | Renders image & textField elements     | The image with altText and textFields are confirmed on the screen                                                                                                                               | Img with altText 'SweetLedger Logo' and two textFields for username and email are validated                                                    | Img is confirmed with altText and both textFields are found on screen | Test Passes |
| ForgotPas<br>s-004 | Renders initial buttons                | The elements with role buttons are queried by their names respectively, 'Next', 'Clear', and 'Back to Login'                                                                                    | All three elements should be found with their names as attributes to each                                                                      | The elements are confirmed in the initial render of the screen        | Test Passes |
| ForgotPas<br>s-005 | Renders secondary elements & navigates | Enters 'something' into all proceeding textboxes, traversing each page by clicking the next button. Then confirming end of page by checking for 'Password has been successfully reset!' prompt. | The page should return the prompt 'Password has been successfully reset!' and the test returns with all expected elements from page navigation | The screen responds that the password has been reset.                 | Test Passes |
| Newuser-<br>001    | Renders New user page                  | Renders New User page and ensures that the page renders with its unique text element 'Create New Account'                                                                                       | Test ensures that render command has passed, and expects that it can find the unique text element                                              | The page is rendered and the text element is found on the page        | Test Passes |
| Newuser-<br>002    | Renders New User text elements         | Parses through all text elements on the New User Page and ensures that they are present in expected parameters                                                                                  | Finds heading, first and last name, address, date of birth, email, username, password, submit request, Clear, and                              | All elements are found and rendered                                   | Test Passes |

|             |                                      |                                                                                                                                                        |                                                                                                                               |                                                             |             |
|-------------|--------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------|-------------|
|             |                                      |                                                                                                                                                        | Back to Login text elements                                                                                                   |                                                             |             |
| Newuser-003 | Renders New user input elements      | Ensures that all textField elements are rendered on this screen and ready for user input                                                               | Expects first name, last name, address, date of birth, email, username, and password input fields to be present on the screen | All elements are present on the screen and ready for input  | Test Passes |
| Newuser-004 | Renders New user clickable elements  | Ensures that all button or link elements are present on the screen for the user to submit data.                                                        | Expects the Submit Request and Clear buttons Back to Login link to be present on the screen                                   | All elements are present on the screen                      | Test Passes |
| Admin-001   | Renders Administrator Page           | Ensures that the page renders properly and finds any elements to ensure that.                                                                          | The page renders with no error and extracts an element to verify that                                                         | Page renders without crashing and renders elements properly | Test Passes |
| Admin-002   | Renders the Header of the admin page | Collects the elements for admin header and user section, as well as dashboard, account mgmt, chart of account, event log, and logout buttons           | The header and user section containers, admin account label, and all buttons are confirmed for presence on the render         | All items are returned as expected                          | Test Passes |
| Admin-003   | Does not Render financial ratios     | Ensures that the label for financial ratios is not rendered on the page                                                                                | Financial ratios are confirmed to be not found on the screen                                                                  | No financial ratio elements found on the page               | Test Passes |
| Admin-004   | Renders admin service navigation     | Ensures that the Access Service buttons are rendered with count 3, and the services grid along with the grid item descriptions are present on the page | The services grid should contain 3 items, and the elements should be found on the render                                      | All items are accounted for and found in the render         | Test Passes |
| Manager-001 | Renders Manager Page                 | Ensures that the page renders properly and finds any elements to ensure that.                                                                          | The page renders with no error and extracts an element to verify that                                                         | Page renders without crashing and renders elements properly | Test Passes |
| Manager-002 | Renders the Header of the            | Collects the elements for Manager header and user section, as                                                                                          | The header and user section containers, Manager account                                                                       | All items are returned as expected                          | Test Passes |

|                |                                           |                                                                                                                                                        |                                                                                                                            |                                                                                  |             |
|----------------|-------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------|-------------|
|                | manager page                              | well as dashboard, account mgmt, chart of account, event log, and logout buttons                                                                       | label, and all buttons are confirmed for presence on the render                                                            |                                                                                  |             |
| Manager-003    | Render financial ratios                   | Ensures that the label for financial ratios is rendered on the page and the grid element is found                                                      | Financial ratios are confirmed found on the screen                                                                         | financial ratio elements found on the page along with the grid holding the items | Test Passes |
| Manager-004    | Renders manager service navigation        | Ensures that the Access Service buttons are rendered with count 7, and the services grid along with the grid item descriptions are present on the page | The services grid should contain 7 items, and the elements should be found on the render                                   | All items are accounted for and found in the render                              | Test Passes |
| Accountant-001 | Renders Accountant Page                   | Ensures that the page renders properly and finds any elements to ensure that.                                                                          | The page renders with no error and extracts an element to verify that                                                      | Page renders without crashing and renders elements properly                      | Test Passes |
| Accountant-002 | Renders the Header of the Accountant page | Collects the elements for Accountant header and user section, as well as dashboard, account mgmt, chart of account, event log, and logout buttons      | The header and user section containers, Accountant account label, and all buttons are confirmed for presence on the render | All items are returned as expected                                               | Test Passes |
| Accountant-003 | Render financial ratios                   | Ensures that the label for financial ratios is rendered on the page and the grid element is found                                                      | Financial ratios are confirmed found on the screen                                                                         | financial ratio elements found on the page along with the grid holding the items | Test Passes |
| Accountant-004 | Renders Accountant service navigation     | Ensures that the Access Service buttons are rendered with count 5, and the services grid along with the grid item descriptions are present on the page | The services grid should contain 5 items, and the elements should be found on the render                                   | All items are accounted for and found in the render                              | Test Passes |
| AcctMgmt-001   | Renders the Account                       | Administrator screen is rendered then there is a navigation to the                                                                                     | After navigation from admin, there is a brief delay to                                                                     | The page successfully navigates and                                              | Test Passes |

|              |                                                   |                                                                                                                                                                                                                                       |                                                                                                                                                                                                  |                                                                                                       |             |
|--------------|---------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------|-------------|
|              | Management Page                                   | account management page for data-passing. The test then gets elements from after data is fetched to the page                                                                                                                          | allow the useEffects to render all components. Then the Administrator Account Management header and fetched user management text is expected.                                                    | fetches relevant data for the render, and header as well as user management text is found on the page |             |
| AcctMgmt-002 | Renders user management table components          | Finds the buttons for user management, 'Create New User', All 'Edit', 'Deactivate', 'Suspend, and 'Email' Buttons. Logs amount of each. Gets all page's table headers and ensures the first 4 match the user management's headers     | All buttons should render after the useEffect, and each return 17 (currently). The table headers, 'Username', 'Email', 'Role', and 'Status' should be found in the first 4 table header elements | All buttons and amounts are returned as intended, and the first first table headers are a match       | Test Passes |
| AcctMgmt-003 | Render Account Management TextFields & Comboboxes | 'Account Name', 'account Number', 'description', 'category', 'subcategory', 'initial Balance', 'debit', 'credit', 'balance', 'user Id', 'order', 'comment' textfields are ensured in the render and both combobox fields are verified | All textfields should be validated in the render after useEffect hook has fetched the table in the screen.                                                                                       | The screen finds all input elements for account creation                                              | Test Passes |
| AcctMgmt-004 | Renders Account Management Text and Button        | Finds the Text Label and Button elements inherent to a post-fetch render of the account management feature of the account management screen                                                                                           | Find a text element matching 'Existing Accounts' and find a button element with name 'Add Account'                                                                                               | The test is able to find both elements matching each name                                             | Test Passes |
| AcctMgmt-005 | Renders Account Management                        | Ensures the Account Management container is present                                                                                                                                                                                   | Expects each header is found in the table for account                                                                                                                                            | All headers are found in the screen in order                                                          | Test Passes |

|                 |                                              |                                                                                                                                                                                                     |                                                                                                                                                                                                    |                                                                                                |             |
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|                 | Table Headers                                | with table headers, 'Account Number', 'Account Name', 'Account Type', 'Subcategory', 'Balance', 'Created By', 'Date Created', and 'Comments'.                                                       | management in order                                                                                                                                                                                |                                                                                                |             |
| ChartofAcct-001 | Renders Chart of Accounts with no Issue      | Ensures that the render of Chart of Accounts from the Administrator's screen proceeds with no issues                                                                                                | Render the Administrator screen then click the button to access render to Chart of Accounts with a brief delay. Finally, ensure there are elements present unique to the Chart of Accounts screen. | The navigation proceeds well and now the fully rendered screen is accessible with fetched data | Test Passes |
| ChartofAcct-002 | Renders the header for Chart of Accounts     | Confirms presence of all heading elements as well as 'View All Accounts Report', 'Email Admin or Manager', and '?' Buttons within the header of the screen.                                         | All headings are expected to be length 3, and all 3 buttons are expected to be rendered in the page.                                                                                               | All headings are length 3 and 3 types of buttons are rendered on the page.                     | Test Passes |
| ChartofAcct-003 | Render search and filter accounts section    | Ensures the header of name 'Search and Filter Accounts', combobox, TextField, Search Button, and Text element 'Select a filter type and enter a search term to find accounts.' are present on page. | All elements should match the criteria for finding each, confirming that all elements of the search and filter accounts section are present                                                        | All elements of the Search and filter accounts section are present.                            | Test Passes |
| ChartofAcct-004 | Render text and button for chart of accounts | For the chart of accounts section of the page, 'Add Accounts' Button and 'Manage your accounts here.' text element must be present                                                                  | Test should find both text and button elements given the criteria                                                                                                                                  | Test finds both elements unique to the section                                                 | Test Passes |



|                     |                                                                |                                                                                                                                                                                                          |                                                                                                                                                                                 |                                                                                                    |             |
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| ChartofAc<br>ct-005 | Render Table<br>for Chart of<br>Accounts                       | Ensures that the table<br>element is rendered<br>and present after<br>useEffects render                                                                                                                  | The test finds a<br>valid table element                                                                                                                                         | A table element is<br>found                                                                        | Test Passes |
| ChartofAc<br>ct-006 | Ensure<br>multiple<br>rendered<br>rows in chart<br>of accounts | Validates that there<br>are multiple rows<br>within the rendered<br>table                                                                                                                                | The test creates an<br>array of row<br>elements in the table<br>and passes as long<br>as it is not zero.                                                                        | The test confirms<br>the list is greater<br>than 0.                                                | Test Passes |
| ChartofAc<br>ct-007 | Render all<br>chart of<br>accounts<br>headers                  | 'Select to Edit',<br>'Account Number',<br>'Account Name',<br>'Account Type',<br>'Subcategory',<br>'Balance', 'Created<br>By', 'Date Created',<br>'Comments' all are<br>Table header names<br>to be found | Creates an array of<br>all table headers and<br>finds each of the 9<br>headers by name in<br>order by render.                                                                   | All 9 headers are<br>found in order on<br>the screen                                               | Test Passes |
| Eventlog-<br>001    | Renders the<br>Eventlog<br>page                                | Ensures that the<br>Eventlog is rendered<br>with no issues from<br>the administrator<br>dashboard to allow<br>for proper<br>data-passing.                                                                | Renders the Admin<br>Dashboard, then<br>clicks the button to<br>access eventlog<br>screen render. After<br>a brief delay, a<br>unique text element<br>is found on the<br>screen | The unique text<br>element 'View all<br>account changes,<br>including before<br>and after states.' | Test Passes |
| Eventlog-<br>002    | Renders the<br>major<br>elements of<br>the Eventlog            | Finds all text<br>elements saying<br>'Event Log' and the<br>table element<br>containing the<br>pertinent information                                                                                     | Create an array of<br>Event Log elements<br>and confirm the list<br>to be greater than or<br>equal to 1. Expects<br>the table element to<br>be present on<br>rendered page.     | Text elements are<br>found, and the<br>table element is<br>found on the<br>screen.                 | Test Passes |
| Eventlog-<br>003    | Renders the<br>table headers                                   | Creates an array of all<br>table headers on<br>screen and checks<br>each by name in the<br>rendered order                                                                                                | Should find every<br>column header<br>element in order<br>according to each<br>name.                                                                                            | All headers are<br>found and present<br>on the page                                                | Test Passes |
| Eventlog-<br>004    | Renders the<br>table rows                                      | Create an array of all<br>table rows and checks<br>that there are multiple<br>rows containing<br>information                                                                                             | The array must<br>return a length<br>greater than or equal<br>to 1                                                                                                              | Confirms there<br>are more than 0<br>rows and prints<br>current count<br>(125)                     | Test Passes |

| Test Case | Purpose                                                                          | Steps                                                                                                   | Expected Result<br>(Running validators in test for all these elements for test pass)                                                                                                         | Actual Result                       |
|-----------|----------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------|
| E2E-001   | Navigate Admin Login to User Management and utilize admin specific functions     | Login, access user management service, access admin specific functions. Return to dashboard then logout | Initialize page, detect successful admin login, access user management service, detect 'activate', 'suspend', 'email' & 'edit' buttons. Return to dashboard. Confirm login validity. Logout. | Fail lacks elements within page *** |
| E2E-002   | Navigate Manager login to User Management and utilize manager specific functions | Login as Manager, access user management service, view all working users, return dashboard then logout  | Initialized                                                                                                                                                                                  |                                     |
| E2E-003   | Navigate accountant login to user management and view fellow users               |                                                                                                         | **                                                                                                                                                                                           | ***                                 |
| E2E-004   | Navigate & Add Ledger entries as Admin                                           |                                                                                                         | **                                                                                                                                                                                           | ***                                 |
| E2E-005   | Navigate & Add Ledger entries as Manager                                         |                                                                                                         | **                                                                                                                                                                                           | ***                                 |
| E2E-006   | Navigate & Add Ledger entries as Accountant                                      |                                                                                                         |                                                                                                                                                                                              |                                     |

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|--|----------------------------------------------|------------------------------------------------|---------------------------------------------|---------------------------------------|
|  | Dynamically<br>adds user to the<br>user list | Verify matching<br>text element on<br>the page | Match<br>keystrokes for<br>user add to page | Fail lacks user<br>entry to user list |
|--|----------------------------------------------|------------------------------------------------|---------------------------------------------|---------------------------------------|